Statistical Design and Models for Evaluation Studies (37495)

Lecture 3 - Diagnostics: assumptions, transformations

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1. Lecture outline

Topics:

- Checking assumptions
- Example 1
- Example 2
- Example 3
- Variance-stabilising transformations
- Box-Cox transformation
- Non-parametric tests

Checking assumptions

- ANOVA works if assumptions are satisfied
- Usual assumption:

$$\boxed{\epsilon_{ij} \overset{i.i.d}{\sim} N(0, \sigma^2)}$$

1. normality
2. independence
3. homogeneity of variance

How do we check the assumptions?

$y_{ij} = \mu_i + \varepsilon_{ij}$

Checking assumptions based on estimated residuals

$$\hat{\epsilon}_{ij} = y_{ij} - \hat{\mu}_i, \text{ for } i = 1, \ldots, a \text{ and } j = 1, \ldots, n$$

1. Normality: qq-plot (or pp-plot); various normality tests
2. Independence: scatter plot of residual; Durbin Watson test
3. Homogeneity of variance: all treatment groups should have residuals approximately equally spread out around 0

Example 1

There are various methods available to estimate the peak discharge from a watershed. Four of these methods were compared. Since there were four methods, we have $a = 4$. Each of the methods was used six times on the watershed and the data are the discharge volumes in cubic feet per second. In this experiment each treatment has $n = 6$ replications.

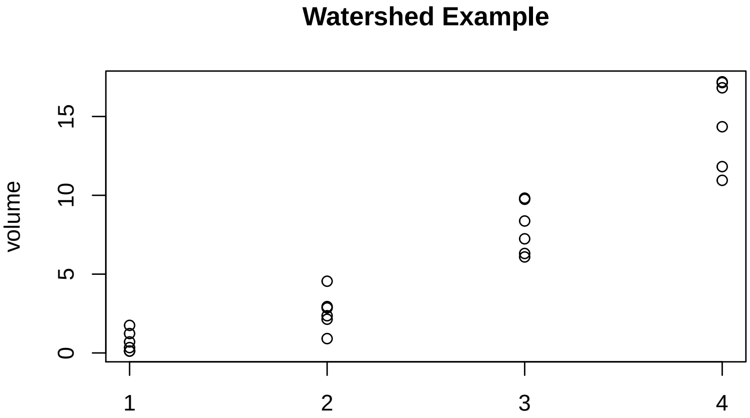
Method 1	**Method 2**	**Method 3**	**Method 4**
0.34	0.91	6.31	17.15
0.12	2.94	8.37	11.82
1.23	2.14	9.75	10.95
0.70	2.36	6.09	17.20
1.75	2.86	9.82	14.35
0.12	4.55	7.24	16.82

Example 1 in R

```
# Remove all objects.
rm(list=ls())
#
# Setup data into data frame.
data1 <- c(0.34,1,0.12,1,1.23,1,0.7,1,1.75,1,0.12,1,0.91,2,2.94,
 2,2.14,2,2.36,2,2.86,2,4.55,2,6.31,3,8.37,3,9.75,3,6.09,3,9.82,
 3,7.24,3,17.15,4,11.82,4,10.95,4,17.2,4,14.35,4,16.82,4)
#response and treatment data
watershed <- data.frame(data1[seq(from=1, to=47, by=2)],
    as.factor(data1[seq(from=2, to=48, by=2)]))
#save in data frame
names(watershed) <- c("volume", "method")
#change names of data frame components
```

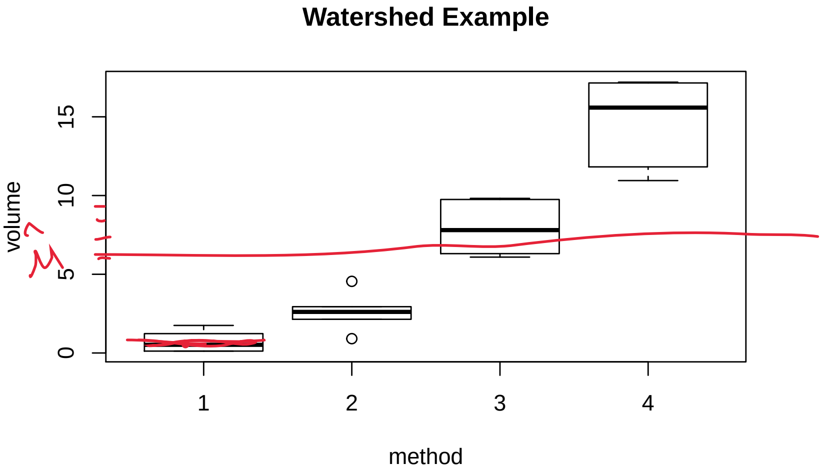
Example 1 in R

```
#Plot data
stripchart(volume~method, vertical=TRUE, pch=1, data=watershed,
    main="Watershed Example")
```



Example 1 in R

```
boxplot(volume~method, data=watershed, main="Watershed Example")
```



$\frac{MSTr}{MSR}$

To obtain the ANOVA output we run the code below.

SS/Df

```
fit.watershed <- aov(volume~method, data=watershed)
summary(fit.watershed)
#               Df Sum Sq Mean Sq F value    Pr(>F)
# method         3  708.3   236.1   76.07 4.11e-11 ***
# Residuals     20   62.1     3.1
# ---
# Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1#
```

$a(n-1)$

$P(F_{3,20} > 76.07) < 0.05$

⇒ there is a significant difference in at least one mean in comparison to the others. Next week we introduce tools to detect which mean/means is/are different

Example 1 in R

Now we obtain regression coefficients and estimates of μ, μ_i and τ_i. $y_{ij} = \mu_i + \varepsilon_{ij}$

```
# Obtain regression coefficients (method 1 as reference).
coef(fit.watershed)
#(Intercept)     method2      method3      method4
#   0.710000    1.916667     7.220000    14.005000
```

$\hat{\mu}_1$

$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$

```
# Obtain least-squares estimates of mu, mu_i and tau_i.
mu_hat <- mean(watershed$volume) #mean across samples
mu_hat
#[1] 6.495417
```

$\hat{\mu} = \bar{y}_{..}$

$\hat{\mu}_i$, $\hat{\mu}$, $\hat{\tau}_i$

```
mu_hat_i <- tapply(watershed$volume, watershed$method, mean)
mu_hat_i
#         1          2          3          4
#0.710000  2.626667  7.930000 14.715000
```

$\hat{\mu}_i$

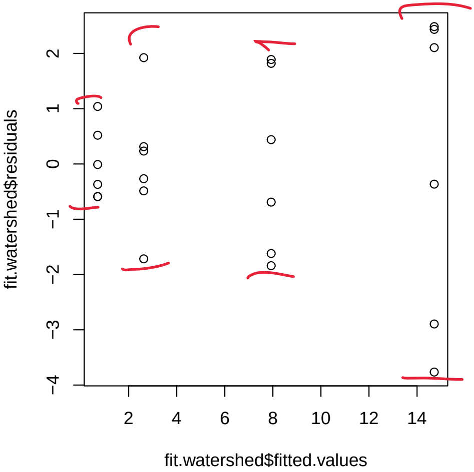
```
tau_hat_i <- mu_hat_i-mu_hat
tau_hat_i
#          1          2          3          4
# -5.785417 -3.868750  1.434583  8.219583
```

$\hat{\mu}_i - \hat{\mu}$

Example 1 in R

Now we look at some diagnostic plots of the aov object.

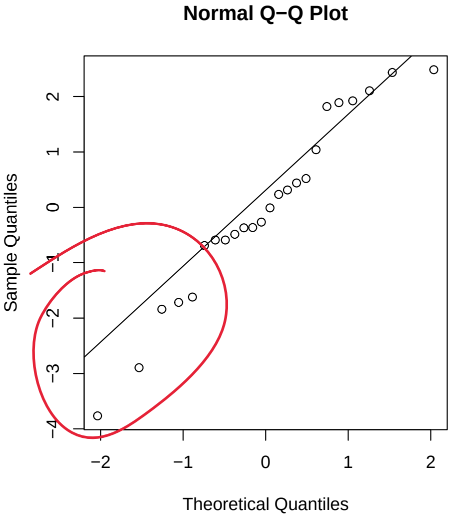
```
plot(fit.watershed$residuals~fit.watershed$fitted.values)
```



Example 1 in R

```
qqnorm(fit.watershed$residuals)
qqline(fit.watershed$residuals)
```

$n = 24$



Example 1 in R

H_0: residuals normaly distributed

Apply some formal normality tests.

```
res.watershed <- residuals(fit.watershed)
# Shapiro-Wilk test.
shapiro.test(res.watershed)
# data:  res.watershed
# W = 0.95701, p-value = 0.3814
#
# Kolmogorov-Smirnov test.
ks.test(res.watershed, "pnorm" , mean=mean(res.watershed),
    sd=sd(res.watershed))
#  One-sample Kolmogorov-Smirnov test
#
# data:  res.watershed
# D = 0.12892, p-value = 0.7734
# alternative hypothesis: two-sided
```

$> 0{,}05$

> 0.05

```
# Cramer-von Mises test.
library(nortest)
# install.packages("nortest") #install if necessary
nortest::cvm.test(res.watershed)
# Cramer-von Mises normality test
# data:  res.watershed
# W = 0.053671, p-value = 0.4432   >0,05
#
# Anderson-Darling test
nortest::ad.test(res.watershed)
# Anderson-Darling normality test
# data:  res.watershed
# A = 0.36391, p-value = 0.4106   >0,05
```

Testing for equal variance assumption.

```
library(lawstat)
# install.packages("lawstat") #install if necessary
levene.test(watershed$volume, watershed$method,
    location="median", correction.method="zero.correction")
#     Modified robust Brown-Forsythe Levene-type test based on the ab
#     deviations from the median with modified structural zero remova
#     method and correction factor
#
# data:  watershed$volume
# Test Statistic = 6.8867, p-value = 0.003436
```

< 0.05

Evidence to reject the equal variance assumption.

Example 2

Larry was interested in determining whether the amount of time (y_{ij}) he had to wait at a particular pedestrian crossing depended on the number of times he/she pushed the button. In this case the response variable is the amount of time that the pedestrian waited at the crossing (in seconds) and the treatment variable is the number of times that he/she pressed the button, once, twice, three times, or not at all.

Observe that these data do not have the same number of observations for each treatment. On this account, the calculation of the parameters estimates and the ANOVA table is slightly different in this case, but the theory is the same.

ANOVA table

Source of Variation	Degrees of freedom	Sum of squares	Mean Square	F-statistic
Treatment	$a-1$	$\sum_{i=1}^{a} n_i(\overline{y}_{i.} - \overline{y}_{..})^2$	$MS_{\text{Tr}} = \frac{ss_{\text{Tr}}}{a-1}$	$f_0 = \frac{MS_{\text{Tr}}}{MS_{\text{Er}}}$
Error	$N-a$	$\sum_{i=1}^{a} \sum_{j=1}^{n_i}(y_{ij} - \overline{y}_{i.})^2$	$MS_{\text{Er}} = \frac{ss_{\text{Er}}}{N-1}$	
Total	$N-1$	$\sum_{i=1}^{a} \sum_{j=1}^{n_i}(y_{ij} - \overline{y}_{..})^2$		

Note: n_i is the sample size of the ith treatment and $N = \sum_{i=1}^{a} n_i$ is the total number of observations.

$n \to n_i$

Example 2

Number of Pushes			
0	**1**	**2**	**3**
38.14	38.28	38.17	38.14
38.20	37.17	38.13	38.30
38.31	38.08	38.16	38.21
38.14	38.25	38.30	38.04
38.29	38.18	38.34	38.37
38.17	38.03	38.34	
38.20	37.95	38.17	
	38.26	38.18	
	38.30	38.09	
	38.21	38.06	

7 10 10 5

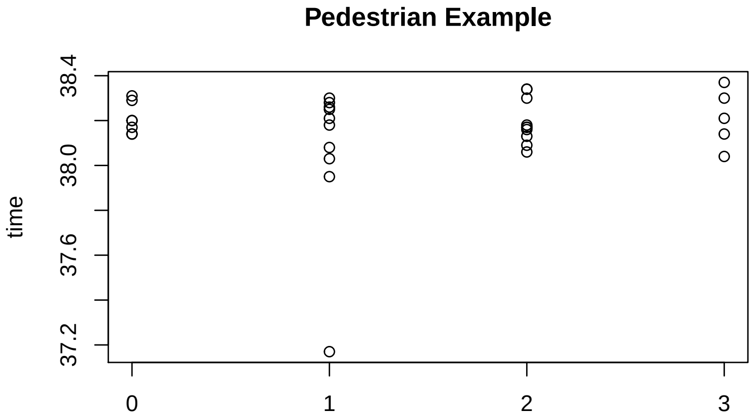
N

Example 2 in R

```
# Setup data
data1 <- c(38.14,0,38.2,0,38.31,0, 38.14,0,38.29,0,38.17,0,38.2,
 0,38.28,1,37.17,1,38.08,1,38.25,1,38.18,1,38.03,1,37.95,1,38.26,
 1,38.3,1,38.21,1,38.17,2,38.13,2,38.16,2,38.3,2,38.34,2,38.34,2,
 38.17,2,38.18,2,38.09,2,38.06,2,38.14,3,38.3,3,38.21,3,38.04,3,
 38.37,3) #response and treatment data
pedestrian <- data.frame(data1[seq(from=1, to=length(data1)-1,
by=2)], as.factor(data1[seq(from=2, to=length(data1), by=2)]))
#save in data frame
names(pedestrian) <- c("time", "pushes")
#change names of data frame components
```

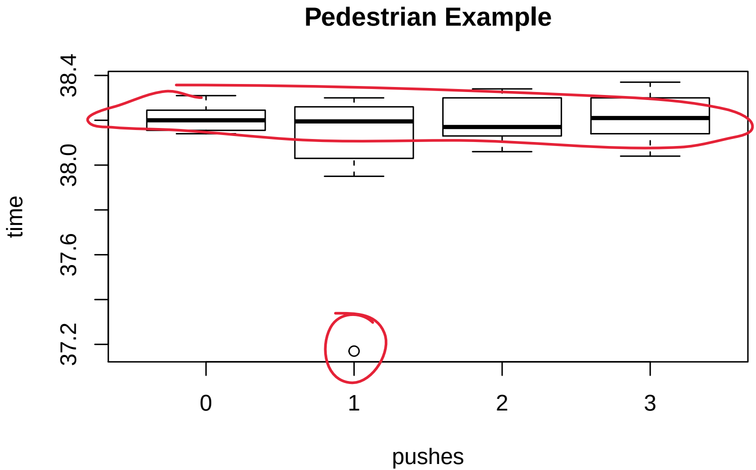
Example 2 in R

```
stripchart(time~pushes, vertical=TRUE, pch=1, data=pedestrian,
    main="Pedestrian Example")
```



Example 2 in R

```
boxplot(time~pushes, data=pedestrian, main="Pedestrian Example")
```



```
# Fit ANOVA model
fit.pedestrian <- aov(time~pushes, data=pedestrian)
summary(fit.pedestrian) #ANOVA table
#                Df Sum Sq Mean Sq F value Pr(>F)
# pushes          3 0.1198 0.03993   0.926  0.441
# Residuals      28 1.2080 0.04314
```

> 0.05

There is no significant difference in means.

$H_0: \mu_0 = \mu_1 = \mu_2 = \mu_3$

Example 2 in R

Now we obtain regression coefficients and estimates of μ, μ_i and τ_i.

```
# Obtain regression coefficients (method 1 as reference).
coef(fit.pedestrian)
# (Intercept)      pushes1      pushes2      pushes3
#38.207142857 -0.136142857 -0.013142857  0.004857143
#
mu_hat <- mean(pedestrian$time)
```

μ

```
mu_hat
#[1] 38.16125
mu_hat_i <- tapply(pedestrian$time, pedestrian$pushes, mean)
```

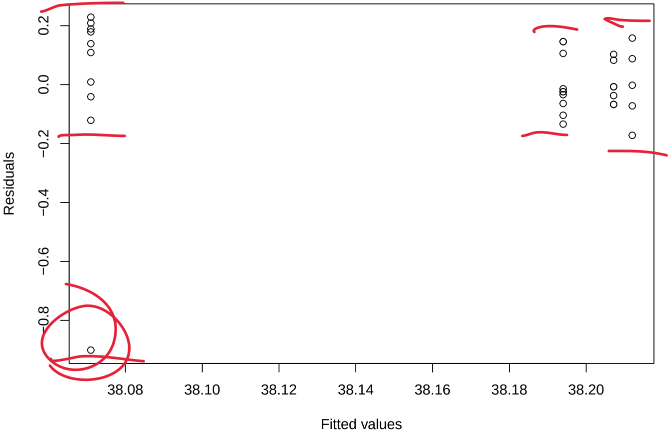
μ_i

```
mu_hat_i
#        0        1        2        3
# 38.20714 38.07100 38.19400 38.21200
#
tau_hat_i <- mu_hat_i-mu_hat
tau_hat_i
#           0           1           2           3
# 0.04589286 -0.09025000  0.03275000  0.05075000
```

Example 2 in R

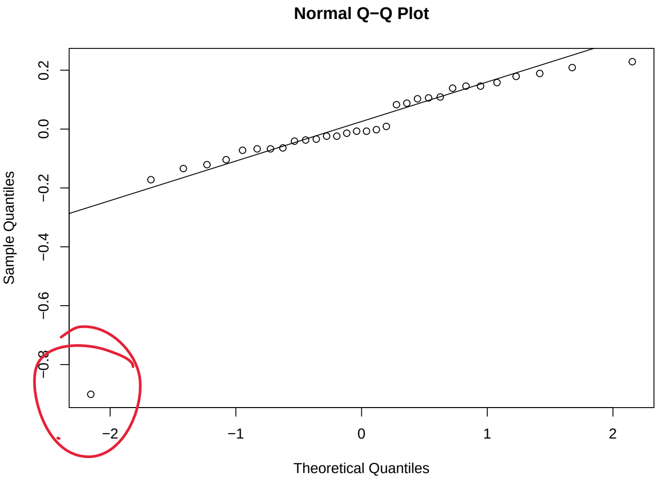
Finally some diagnostic plots

```
res.pedestrian <- residuals(fit.pedestrian)
plot(res.pedestrian~fit.pedestrian$fitted.values,
xlab="Fitted values",ylab="Residuals")
```



Example 2 in R

```
qqnorm(res.pedestrian)
qqline(res.pedestrian)
```



Example 2 in R

Apply some formal normality tests.

```
# Shapiro-Wilk test
shapiro.test(res.pedestrian)
#  Shapiro-Wilk normality test
#
# data:  res.pedestrian
# W = 0.70253, p-value = 9.488e-07
#
# Kolmogorov-Smirnov test
ks.test(res.pedestrian, "pnorm" , mean=mean(res.watershed),
    sd=sd(res.watershed))
#  One-sample Kolmogorov-Smirnov test
#
# data:  res.pedestrian
# D = 0.44457, p-value = 6.42e-06
# alternative hypothesis: two-sided
# Warning message:
# In ks.test(res.pedestrian, "pnorm", mean = mean(res.watershed),
# ties should not be present for the Kolmogorov-Smirnov test
```

Example 2 in R

```
# Cramer-von Mises test.
library(nortest) # install.packages("nortest")
nortest::cvm.test(res.pedestrian)
#  Cramer-von Mises normality test
#
# data:  res.pedestrian
# W = 0.28461, p-value = 0.0004462
#
# Anderson-Darling test.
nortest::ad.test(res.pedestrian)
#  Anderson-Darling normality test
#
# data:  res.pedestrian
# A = 2.0021, p-value = 3.196e-05
```

Normality fails on all counts, but this could be due to the massive outlier → remove this points and redo the analysis (details omitted). Note also that the Kolmogorov-Smirnov test should not be used in this case.

Example 2 in R

Testing for equal variance assumption.

```
library(lawstat)
# install.packages("lawstat") #install if necessary
levene.test(pedestrian[,"time"],  pedestrian[,"pushes"],
    location="median", correction.method="zero.correction")
#    Modified robust Brown-Forsythe Levene-type test based on the ab
#    deviations from the median with modified structural zero remova
#    method and correction factor
#
# data:  pedestrian[, "time"]
# Test Statistic = 1.0685, p-value = 0.381
```

Fail to reject the equal variance assumption.

Variance-stabilising transformations

- If Levene's test is significant then it is sometimes possible to apply a function to the data in order to reach an approximately constant variance.
  $\Rightarrow$ Variance-stabilizing transformatons.
- Often such transformation is a power transformation and it is determined empirically based on the data, although care must be taken to ensure that the transformation makes sense in the context of the experiment
- To make transformations in R, we use the command `newvar <- funct(oldvar)` where `funct` refers to any mathematical functions that are defined in R.

Example 1 revisited

Recall that we had a problem with the variance ~~data~~ in the watershed data. Let's try take the square root of the response variable.

```
watershed$volume.sqrt <- sqrt(watershed$volume)
# Levene's test of equal variances on transformed response
levene.test(watershed$volume.sqrt,  watershed$method,
  location="median", correction.method="zero.correction")
#    Modified robust Brown-Forsythe Levene-type test based on the ab
#    deviations from the median with modified structural zero remova
#    method and correction factor
#
# data:  watershed$volume.sqrt
# Test Statistic = 0.36495, p-value = 0.7792
```

> 0.05

We see that the square-root transformation has stablised the variance acrosss the samples. We now re-do the ANOVA and check the residuals for assumptions.

Example 1 revisited

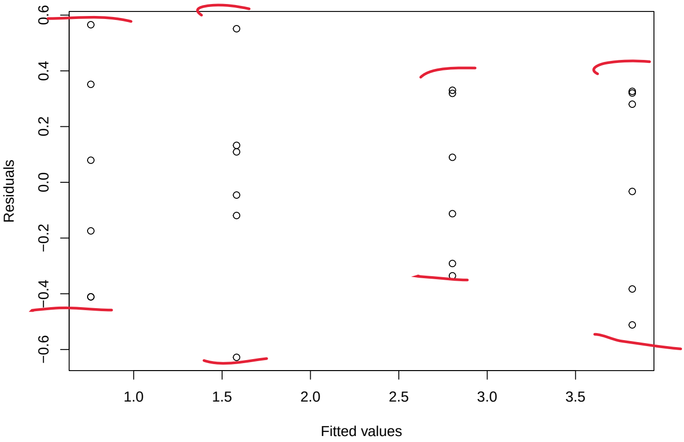
Recall that we had a problem with the variance data in the watershed data. Let's try take the square root of the response variable.

```
#Re-fit model on transformed data
fit.watershed.sqrt <- aov(volume.sqrt~method, data=watershed)
summary(fit.watershed.sqrt)
#               Df Sum Sq Mean Sq F value  Pr(>F)
# method         3  32.68  10.895   81.05 2.3e-11 ***
# Residuals     20   2.69   0.134
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1#
```

We still have a significant result from the F-test. Next we look at some diagnostic plots.

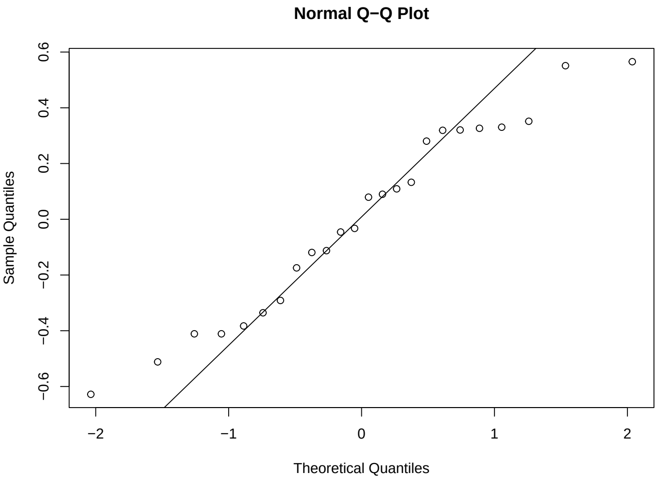
Example 1 revisited

```
res.watershed.sqrt <- residuals(fit.watershed.sqrt)
plot(res.watershed.sqrt~fit.watershed.sqrt$fitted.values,
xlab="Fitted values",ylab="Residuals")
```



Example 1 revisited

```
qqnorm(res.watershed.sqrt)
qqline(res.watershed.sqrt)
```



Example 1 revisited

Formal normality tests

```
shapiro.test(res.watershed.sqrt)
#  Shapiro-Wilk normality test
#
# data:  res.watershed.sqrt
# W = 0.95877, p-value = 0.4141
#
# Kolmogorov-Smirnov test.
ks.test(res.watershed.sqrt, "pnorm" ,
 mean=mean(res.watershed.sqrt), sd=sd(res.watershed.sqrt))
#  One-sample Kolmogorov-Smirnov test
#
# data:  res.watershed.sqrt
# D = 0.12726, p-value = 0.7863
# alternative hypothesis: two-sided
```

Example 1 revisited

```
# Cramer-von Mises normality test
nortest::cvm.test(res.watershed.sqrt)
#  Cramer-von Mises normality test
#
# data:  res.watershed.sqrt
# W = 0.053365, p-value = 0.44744
#
# Anderson-Darling normality test.
nortest::ad.test(res.watershed.sqrt)
#  Anderson-Darling normality test
#
# data:  res.watershed.sqrt
# A = 0.3595, p-value = 0.4207
```

All normality tests fail to reject the normality assumption.

Box-Cox transformation

- In practice, the data are often transformed using logarithm or square root functions
- These two transformations are in fact special cases of the so-called Box–Cox transformation
- The procedure considers a family of transformations and simultaneously estimates the best parameter from the perspective of making variances homogeneous among treatment groups
- More precisely, the additional parameter λ changes the response variable y as

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda \left(\exp \left[\sum_{i=1}^{N} \ln(y_i)/N \right] \right)^{\lambda - 1}} & \text{for } \lambda \neq 0; \\ \exp \left[\sum_{i=1}^{N} y_i / N \right] \times \ln(y) & \text{for } \lambda = 0. \end{cases}$$

Box-Cox transformation

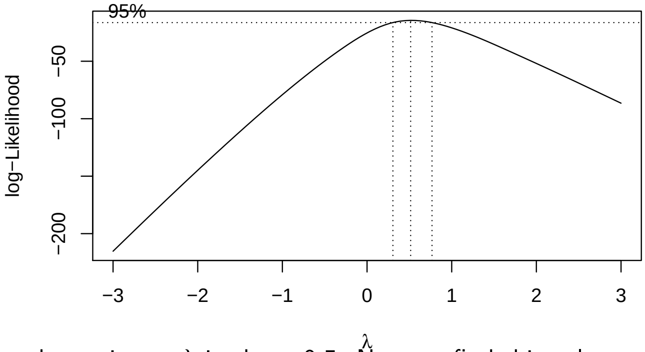
- As λ tends to 0, the term $(y^\lambda - 1)/\lambda$ tends to $\ln(y)$
- Often easier to use easily explainable transformations such as $\lambda = -1, 0, 0.5, 2$
- Note that the response variable must be positive

Example 1 revisited

Consider the watershed data again. In the previous example we chose to take the square root transformation, probably after a deal of trial and error. If we were not sure about an appropriate transformation, we could use a Box–Cox transformation where the parameter λ is estimated for us.

Example 1 revisited

```
library(MASS)
# Log-likelihood plot.
boxcox(volume~method, data=watershed, lambda=seq(-3, 3, length=100))
boxcox(fit.watershed, lambda=seq(-3, 3, length=100))
#works also on aov object
```



We see that the optimum λ is about 0.5. Now we find this value, transform the data and see if variance has been stabilised.

Example 1 revisited

```
parlist <- boxcox(fit.watershed, lambda=seq(-3, 3, length=100),
plotit=FALSE)# Save optimum lambda.
lambda_max <- parlist$x[which.max(parlist$y)]
lambda_max
# 0.5151515
# Apply Box-Cox transformation and save in data frame.
library(DescTools)
# install.packages("DescTools") #install if necessary
watershed$volume.BC <- BoxCox(watershed$volume, lambda_max)
# Perform Levene's test on transformed data.
levene.test(watershed$volume.BC,  watershed$method,
    location="median", correction.method="zero.correction")
#  Modified robust Brown-Forsythe Levene-type test based on the abso
#  deviations from the median with modified structural zero removal
method and correction factor
#
# data:  watershed$volume.BC
# Test Statistic = 0.32025, p-value = 0.8106
```

Example 1 revisited

Finally we re-do the ANOVA on the Box-Cox transformed data.

```
fit.watershed.BC <- aov(volume.BC~method, data=watershed)
summary(fit.watershed.BC) #ANOVA table
#              Df Sum Sq Mean Sq F value   Pr(>F)
# method        3 136.68   45.56   81.97 2.07e-11 ***
# Residuals    20  11.12    0.56
# ---
# Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Significant result from ANOVA, what about normality?

Example 1 revisited

```
# Shapiro-Wilk test.
res.watershed.BC <- residuals(fit.watershed.BC)
shapiro.test(res.watershed.BC)
#  Shapiro-Wilk normality test
#
# data:  res.watershed.BC
# W = 0.95418, p-value = 0.3328
```

Normality OK.

Non-parametric test

- ANOVA relies on assumptions
- What if assumptions fail?
- Variance stabilising transformations might work, but not always
- We need a more robust test, one which does not require as many assumptions on the distribution of the data
- This is called a distribution free, or nonparametric test

Non-parametric test

The appropriate nonparametric alternative to a one-way ANOVA is the Kruskal-Wallis test. The hypotheses for this test are

$$H_0 : \text{the distributions are the same for all treatments}$$
$$H_1 : \text{the distributions are not the same for all treatments.}$$

In this test, we assign ranks to the observations, and compare the observed ranks that appear in each group to the expected ranks if the null hypothesis were true.

More formally, suppose that the ith set of observations has cumulative distribution function F_i for $i = 1, \ldots, a$ and assume the independence across and within samples. Then we can express our hypotheses as

1 2 3 50

$$H_0 : F_1 = F_2 = \ldots = F_a$$
$$H_1 : \text{Not all } F_i \text{ are the same.}$$

1 2 3 4

Non-parametric test

Under the null hypothesis, we have $n \times a$ observations from one population, so we can assign ranks

$$R_i = \text{sum of ranks for } i\text{-th treatment group}$$

$$\sum_{i=1}^{a} R_i = \frac{N(N+1)}{2} \qquad \text{where } N = na \text{ the total number of observations}$$

$$\mathbb{E}(R_i) = \frac{n}{N} \times \frac{N(N+1)}{2} = \frac{n(N+1)}{2}.$$

The test statistic is based on $(R_i - \mathbb{E}(R_i))^2$ and is

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{a} \left(R_i - \frac{n(N+1)}{2} \right)^2.$$

One can show that H approximately follows χ_1^2-distribution for large n.

Some remarks

- The Kruskal-Wallis test is applicable if the assumptions of normality and constant variance are not satisfied
- If the assumptions are satisfied, this test is less powerful than ANOVA. I.e. this test is less likely to detect differences between treatments (if they exist)
- The Kruskal-Wallis test uses approximation (χ^2_1-distribution) of the the distribution of the test statistic. However, if the sample size in each group is small (less than 5), then one use tables of the exact p-values
- If each treatment group distribution has similar shape then the hypotheses can be ~~res~~-stated in terms of medians

Example 1 revisited

Suppose that instead of performing a variance stabilizing transformation for the watershed data, we apply the Kruskal-Wallis test.

```
kruskal.test(volume~method, data=watershed)
# Kruskal-Wallis rank sum test
#
# data:  volume by method
# Kruskal-Wallis chi-squared = 21.156, df = 3, p-value = 9.771e-05
```

The null hypothesis that the distributions are the same can be rejected.